TECHNICAL REPORT

Combining Label Deconvolution and Frozen Walk-Text Embeddings for Product Classification

A validation-locked, ten-seed study with SAGN/SLE and Correct & Smooth

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Abstract

We study node classification on **ogbn-products** by combining released Label Deconvolution (LD) BERT features, frozen EmbeddingGemma encodings of graph-walk text, and original OGB features. The resulting 1,636-dimensional node representation feeds a four-stage SAGN model with self-label-enhanced training (SLE), followed by Correct & Smooth (C&S). Across ten fresh seeds, the validation-locked v2 pipeline achieves **94.0584% ± 0.0433% validation accuracy** and **87.4984% ± 0.0490% test accuracy** (mean ± sample standard deviation).

1. Dataset and study design

We use the official ogbn-products split and the OGB 1.3.6 evaluator. The dataset contains 2,449,029 nodes, 47 classes, and 61,859,140 undirected product relations, represented after loading as 123,718,280 directional edge records.

Training nodes	Validation nodes	Test nodes
196,615	39,323	2,213,091

V2 uses seeds 10-19 and reports all ten runs without selection. Seeds 0-9 were used during v1 exploration. Sample standard deviations use the n - 1 denominator. **External data: Yes.** [1, 7]

Scope of the result. The reported accuracy belongs to the complete LD + frozen walk-text + raw features + SAGN/SLE + C&S pipeline.

2. Method

2.1 Node representation

Component	Dim.	Source
LD BERT features	768	Released products_sagn/hidden_state.pt
Frozen walk-text features	768	Mean-pooled, L2-normalized EmbeddingGemma encodings
Original OGB features	100	Original ogbn-products node features
Concatenated input	1,636	Order: LD, walk-text, raw

LD features are reused from the released feature tensor; LD is not retrained in this experiment. The concatenated features are then used to train the downstream SAGN/SLE predictor. [2, 3]

2.2 WalkLM and Label Deconvolution: a product example

Example product graph. In an illustrative co-purchase graph, headphones connect to earbuds and a phone case; the earbuds connect to Sony WF-1000XM5 earbuds. Products are nodes, descriptions are attributes, and links indicate co-purchases.

WalkLM: read product text along a graph walk. A graph walk follows connected nodes, for example **Headphones → Earbuds → Sony WF-1000XM5 earbuds**. WalkLM converts the visited products' attributes into text and fine-tunes a language model on these sequences. Its embeddings combine product meaning with graph context. Our implementation uses walk-text with a frozen encoder. [4]

LD (Label Deconvolution): prepare text features for graph aggregation. A GNN mixes the headphones' features with neighboring audio products and a phone case, although connected products may have different categories. LD uses training-label information propagated over the graph to learn adjusted training targets for the text encoder, helping it produce features suited to that later mixing. The headphones' correct category remains unchanged. We reuse released LD-trained BERT embeddings. [3]

The encoder is **google/embeddinggemma-300m**, loaded with AutoModel. Its final hidden states are mean-pooled over non-padding tokens using the attention mask, then L2-normalized. The encoder is frozen. Unlike original WalkLM, this experiment does not fine-tune a language model on graph-walk text; **WalkLM-style** refers only to graph-walk textualization. [4, 6]

2.3 Predictor and post-processing

SAGN/SLE configuration	Value
Feature hops / label hops	2 / 9
Hidden channels / attention heads	512 / 1
Learning rate / batch size	0.001 / 50,000
Dropout values (reported order)	0.5 / 0.4 / 0.2 / 0.5
Epochs across four stages	300 / 100 / 100 / 100
Self-training thresholds	0.95 / 0.95 / 0.999

C&S uses 50 correction iterations and 50 smoothing iterations, correction alpha 0.35, smoothing alpha 0.60, and scale 20. Only training labels enter SAGN/SLE label features and C&S propagation. Validation labels support early stopping and selection; test labels are excluded from propagation. [2, 5, 7]

3. Results and interpretation

All accuracies below are percentages, reported as mean ± sample standard deviation. V1 is exploratory and test-aware; v2 uses the later validation-locked protocol. [7]

Evaluation	Validation (%)	Test (%)
V1 + C&S: exploratory	94.0010 ± 0.0299	87.4753 ± 0.0487
V2 final stage before C&S	93.9204 ± 0.0522	Not evaluated
V2 validation-selected C&S	**94.0584 ± 0.0433**	**87.4984 ± 0.0490**

The source code indexes the four training stages as 0-3; the pre-C&S row refers to stage 3, the fourth and final stage. Test accuracy was not computed for that row.

3.1 Historical leaderboard comparison

The report records the published **GLEM+GIANT+SAGN+SCR** result as **87.37% ± 0.06%** on September 4, 2026. The v2 mean is numerically higher by **0.1284 percentage points**. The other reported comparator, **LD+GIANT+SAGN+SCR**, achieves **87.18% ± 0.04%**. These are historical reference values, not a newly verified leaderboard snapshot. Official eligibility and ranking are determined by OGB.

This comparison does not establish a statistically significant improvement or identify its cause. No v2 walk-text-only ablation was performed, and the full method inherits choices made during earlier test-visible development.

3.2 Validation selection margin

C&S increases mean v2 validation accuracy by approximately **0.1381 percentage points**. The winning C&S setting exceeds the runner-up by only **0.00762912 percentage points**: three additional correctly classified validation nodes per seed on average, or 30 across ten seeds. This small selection margin should be considered when interpreting the selected configuration.

4. Parameter accounting

Component	Parameters
LD bert-base-uncased encoder	109,482,240
LD SAGN/SLE head and aggregation weight	1,154,656
EmbeddingGemma encoder	307,581,696
Final 1,636-input SAGN/SLE predictor	5,010,014
Correct & Smooth	0
Total across listed components	**423,228,606**

This total accounts for all listed upstream and downstream components; it is not a count of parameters jointly trained in the final SAGN run. The two LD rows sum to the published 110,636,896 parameters associated with LD+GIANT+SAGN+SCR. The final SAGN runtime count is recorded in the preserved training logs. [7]

5. Evaluation protocol and historical disclosure

5.1 Earlier test-aware exploration (v1)

V1 compared encoders, feature combinations, SAGN settings, and C&S candidates while test accuracy was visible. The inherited configuration includes two feature hops (K = 2), 512 hidden channels, the epoch schedule, and batch size. These are test-aware historical choices, not settings independently selected by v2 validation.

Recovered history establishes a lower bound of approximately **1,550 epoch-level test reports**: 300 from the final ten-seed SAGN training and about 1,250 from identifiable completed or partial subset-classifier runs. Additional comparisons suggest a larger total, plausibly a few thousand, but the exact count cannot be recovered. The protocol issue is method selection after viewing test accuracy, rather than feeding test labels into the predictor.

5.2 Walk configurations explored during v1

Status	Configuration and preserved outcome
Completed subset screens	2 walks / 3 steps: 256-token and 128-character limits; 4 walks / 4 steps: 160-token and 160-character limits.
Partial subset screen	4 walks / 5 steps, 160-token and 128-character limits: walk-text-only evaluation completed; concatenated-feature evaluation did not.
Failed trials	4 walks / 5 steps and 6 walks / 4 steps with 192-token limits ended during model loading or corpus preparation because of memory pressure.
Submitted feature artifact	Frozen EmbeddingGemma; 4 walks of length 4; 160-token and 160-character limits.

The controlling protocol document distinguishes completed, partial, failed, and proposed trials and records per-encoder outcomes. It should accompany any submission or eligibility review. [7]

5.3 Validation-locked evaluation (v2)

Before v2 execution, preregistration froze seeds 10-19, the inherited model configuration, the 27-member C&S grid, and the selection rule. The grid was inherited from the focused v1 search. V1 selected (correction alpha, smoothing alpha, scale) = **(0.30, 0.65, 25)** using test results; that setting ranks 11th of 27 on v2 validation. V2 instead selected **(0.35, 0.60, 20)** using validation accuracy only.

V2 training masked test labels and reported no test metric. All 27 candidate records declare **test_metrics_accessed: false**. Candidate records and the selected configuration lock were committed before the original final-test command. That guarded command wrote a durable access marker before loading the dataset.

A later public-release reproduction reran the final-only evaluator under a separate reproduction marker after the lock and original result were public. These are the two preserved post-lock v2 test computations; the audit found no evidence of an unguarded v2 test computation.

Interpretation boundary. V2 provides a forward test-isolated evaluation of an inherited method. It does not turn the full development history into a test-independent study or remove v1 selection effects. The controlling disclosure is TUNING_AND_PROTOCOL.md. [7]

6. Reproducibility and artifact checks

The repository pins SAGN and LD revisions, includes the feature-integration patch, records external tensor hashes and sizes, and supplies the EmbeddingGemma reconstruction. Original generation-library versions, internal computation dtype, GPU, and resolved model revision were not captured; they remain unknown. [7]

6.1 Embedding reconstruction check

A deterministic 128-node prefix was reconstructed using the pinned model and settings on an NVIDIA RTX PRO 4500 Blackwell. All **98,304 float32 values** matched the archived tensor rows exactly: mean cosine similarity 1.0, mean L2 difference 0.0, and maximum absolute difference 0.0. This verifies the tested prefix, not a full regeneration of every embedding. The comparison JSON and reconstruction environment are committed.

6.2 Evaluation replay from released predictions

The v2-from-release script downloaded and verified all ten prediction tensors, recomputed all 27 validation candidates, recovered the identical configuration lock, and reproduced test accuracy **0.8749843545 ± 0.0004904255**. This is an evaluation replay from released predictions, not a fresh retraining of all ten models. Its compressed log and SHA-256 are preserved.

The public v2 release includes predictions, hashes, environment records, and logs. Small v1 provenance files are in the repository; large assets are mirrored in an archived release. Each seed log ends with an upstream SAGN NumPy-summary traceback after the final-stage checkpoint was saved. The wrapper verifies that checkpoint before continuing.

7. Conclusion

The complete pipeline achieves **87.4984% ± 0.0490% test accuracy** across ten fresh seeds under the v2 validation-locked protocol. Released predictions and selection artifacts support auditing of the reported arithmetic. The result does not isolate a gain from walk-text or establish an official OGB rank. Any submission should declare external data and retain the full v1 historical disclosure.

References and supporting artifacts

[1] Hu et al. Open Graph Benchmark: Datasets for Machine Learning on Graphs. NeurIPS 2020. Source.

[2] Sun et al. Scalable and Adaptive Graph Neural Networks with Self-Label-Enhanced Training. Source.

[3] Shi et al. Label Deconvolution for Node Representation Learning on Large-scale Attributed Graphs against Learning Bias. Source.

[4] Tan et al. WalkLM: A Uniform Language Model Fine-tuning Framework for Attributed Graph Embedding. NeurIPS 2023. Source.

[5] Huang et al. Combining Label Propagation and Simple Models Out-performs Graph Neural Networks. ICLR 2021. Source.

[6] Google. EmbeddingGemma documentation. Source.

[7] Experiment code, technical report, releases, and controlling tuning/protocol disclosure. Source.

Controlling protocol disclosure · Original technical report